

Mechanical Assembly and Retention Record

A synthetic controlled-drawing extract for testing design-intent retrieval. It is not an actual engineering drawing and contains no fabrication dimensions.

Document ID	Incident	Record type	Status
R3-EV-003	R3 carrier beam - 10/14/2024	Application test evidence	Synthetic / controlled

Synthetic drawing notes

Feature	Test-scenario design statement	Evidence role	Status
Carrier beam / link connection	The carrier beam remains connected to the link by a pinned joint. The keeper retains that pin.	Explains why beam, link, and keeper could be recovered together.	Synthetic scenario fact
Link / yoke interface	The upper link surface rests in an open yoke pocket during the normal supported position.	Defines a possible lift-out or unseating path that must be evaluated from geometry and marks.	Synthetic scenario fact
Keeper function	The keeper is not a positive hold-down over the yoke pocket.	Separates 'keeper present' from 'link positively retained in yoke.'	Synthetic scenario fact
Outage position	When rolls are absent and the balance is raised, the carrier-beam assembly requires either the specified supported state or a rated temporary restraint before hydraulic pressure is removed.	Creates a procedure/design requirement to compare with field execution.	Synthetic scenario fact
Permitted clearances	No acceptance dimensions are included in this extract.	Full drawing and OEM tolerances remain necessary for dimensional conclusions.	Open / request full drawing

Revision and authority

Drawing reference	Revision	Approval	Limitation
R3-RB-ASM-SYN-001	Test Rev A	Synthetic evidence author	For application evaluation only; not valid for field work.

Energy Control Plan - R3 Outage Configuration

A synthetic equipment-specific energy-control plan used to test source, boundary, verification, and configuration reasoning.

Document ID	Incident	Record type	Status
R3-EV-004	R3 carrier beam - 10/14/2024	Application test evidence	Synthetic / controlled

Energy-source boundary

Source ID	Energy / equipment	Planned isolation	Required verification	Status in plan
H-B	B hydraulic system	Shut supply, secure valve, open designated bleed / return path.	Local gauge zero and functional try where applicable.	Defined
H-RB	Top backup roll balance hydraulic branch	Isolate branch before work within the stand.	Confirm no commanded motion and document pressure state.	Defined
G-RB	Gravity - raised roll-balance and attached components	Lower to designated rest position or install rated mechanical blocking/restraint.	Visual and physical verification of support; record block ID.	Defined
M-ROLL	Mechanical support changed by removal of work and backup rolls	Treat removal sequence as a configuration change; verify remaining parts are supported.	Supervisor sign-off before removing hydraulic support.	Defined
E-AUX	Auxiliary electrical controls	Control-power isolation if work enters electrical or motion hazard zone.	Meter test where electrical isolation is applied.	Conditional

Required sequence

- Identify the as-found roll and balance configuration before isolation.
- Establish the safe supported position for gravity-loaded components.
- Apply hydraulic isolation and bleed pressure.
- Verify zero-energy state and positive mechanical support independently.
- Record any departure from the sequence and obtain authorization before work continues.

Execution evidence

This plan states what is required. It does not prove which steps were performed. Compare it with the signed lockout verification, shift log, work scope, photographs, and witness evidence.

Executed Lockout Verification - R3 Outage

A synthetic execution record that should be compared with the Energy Control Plan rather than treated as self-proving.

Document ID	Incident	Record type	Status
R3-EV-005	R3 carrier beam - 10/14/2024	Application test evidence	Synthetic / controlled

Recorded field entries

Entry	Recorded value	Verification / signature	Record status
B hydraulic supply	Isolation valve shut and equipment lock applied.	Initialed - 10/13/2024, 11:58 PM	Completed
B hydraulic bleed	Bleed path opened; local gauge read 0 psi.	Initialed - 10/14/2024, 12:28 AM	Completed
Top backup roll balance position	Raised.	Position noted; no lowering entry	Completed as observation
Gravity control for raised balance / carrier assembly	Mechanical block or rated restraint ID field is blank.	No independent verifier signature	Gap in record
Functional try	No motion on command after hydraulic isolation.	Initialed - Turn 1	Completed
Shift transfer	Open group isolation carried into Turn 2.	Incoming-lead acceptance box unsigned on scanned copy	Gap in record

Record cautions

- A zero-pressure indication does not establish control of gravity energy.
- A blank field is evidence of a documentation gap; it is not proof that the physical control was absent.
- The original paper record and group-box log should be inspected for missing reverse-side or attachment pages.

Shift Turnover Log - Turn 1 to Turn 2

A synthetic contemporaneous log used to test sequence, handoff, and unresolved-condition analysis.

Document ID	Incident	Record type	Status
R3-EV-006	R3 carrier beam - 10/14/2024	Application test evidence	Synthetic / controlled

Turn 1 outgoing entry

Time	Entry	Author role
10/13 10:20 PM	R3 work rolls removed; roll-change equipment clear.	Maintenance shift lead
10/13 11:45 PM	Top and bottom backup rolls removed for outage scope.	Maintenance shift lead
10/14 12:25 AM	Top backup roll balance remains up. B hydraulic system isolated and bled for B-system work.	Energy-isolation lead
10/14 12:40 AM	Carrier-beam support / yoke seating not independently checked after roll removal.	Maintenance shift lead
10/14 12:55 AM	Turnover note: keep personnel clear below upper balance components until day-turn confirms final supported condition.	Maintenance shift lead

Turn 2 incoming entry

Field	Recorded value	Status
Outstanding condition acknowledged	General red-zone restriction acknowledged in verbal handoff.	Synthetic scenario fact
Specific carrier-beam / yoke verification	No entry found.	Gap in record
Formal re-brief before pit work	Morning contractor brief references overhead hazards but not the carrier-beam condition by name.	Synthetic scenario fact

Outage Work Scope and Sequence - R3 Stand

A synthetic work-management record for testing planned-versus-actual configuration and change-control reasoning.

Document ID	Incident	Record type	Status
R3-EV-007	R3 carrier beam - 10/14/2024	Application test evidence	Synthetic / controlled

Approved scope

Step	Planned activity	Interface / hold point
1	Remove work rolls.	Confirm roll-change equipment clear.
2	Remove top and bottom backup rolls.	Confirm balance and associated components in approved supported state.
3	Isolate and bleed B hydraulic system.	Energy-isolation verification before B-system maintenance.
4	Perform mill-stand and hydraulic outage work.	Maintain exclusion zones and scaffold controls.
5	Inspect and restore equipment.	Engineering release before re-energization.

Execution notes

Question	Record finding	Status
Was the listed sequence followed?	Roll removal preceded B-system isolation and bleed.	Synthetic scenario fact
Was the supported-state hold point signed?	No signature or attached checklist is present for Step 2.	Gap in record
Was the carrier beam listed as a separate protected component?	No; it appears only within the general balance-system boundary.	Synthetic scenario fact
Was a field change approved?	No change request is attached.	Open / verify

JJHAC / JSHA - R3 Mill Pit Outage Work

A synthetic pre-job hazard record modeled on the partner document family. It supports hazard-recognition and control-adequacy questions.

Document ID	Incident	Record type	Status
R3-EV-008	R3 carrier beam - 10/14/2024	Application test evidence	Synthetic / controlled

Task and crew

Job / task	Applicable roles	Area	Brief time
R3 pit access and mill-stand outage support	Contractor mechanical crew; site maintenance liaison	R3 roughing mill / mill pit	10/14/2024 - 6:35 AM

Job steps, hazards, and controls

Step	Potential hazard	Recorded control	Assessment
Enter and work from pit / scaffold	Access, fall exposure, dropped objects from above.	Inspect scaffold; barricade work zone; hard hats; keep clear of overhead work.	Addressed generally
Work below mill stand	Suspended or unsecured mill components.	Verify no active crane lift; confirm equipment is under energy control.	Addressed generally
Hydraulic-system outage work	Stored pressure or unexpected motion.	Use ECP/LOTO; verify gauge zero; perform try.	Addressed
Changed roll configuration	Change in support path after roll removal.	No component-specific prompt for carrier beam / link / yoke seating.	Gap in specificity
Conditions change	New task, equipment state, or personnel.	Stop and revise brief before proceeding.	Defined

PPE and tools

Recorded: hard hat, safety glasses with side shields, safety footwear, gloves selected for task, hearing protection, fall protection where required by access condition. PPE does not replace control of falling-object or gravity-energy hazards.

Witness Statement Packet

Two synthetic attributed accounts. The statements preserve observation limits and should not be blended into a single voice.

Document ID	Incident	Record type	Status
R3-EV-009	R3 carrier beam - 10/14/2024	Application test evidence	Synthetic / controlled

Witness W-01 - contractor employee

Prompt	Statement
Position	Walking beside the R3 stand on the operator-side route; not inside the pit.
Observation	Saw the lower beam move slightly at one end, rotate, and then drop into the pit. Did not see a crane hook, rigging, or a person touching it.
Sound	Heard one metallic scrape followed by the impact.
Visibility limit	Could see the beam but not the upper link-to-yoke contact point.
Time confidence	Approximate; based on phone time checked after the event.

Witness W-02 - contractor employee

Prompt	Statement
Position	Several feet behind W-01 on the same walking route.
Observation	Looked toward the stand after hearing a scrape and saw the beam already moving downward. No crane motion was noticed.
Prior condition	Did not inspect the carrier beam or yoke before the event.
Visibility limit	Scaffold and stand structure partially blocked the lower landing area.
Uncertainty	Cannot say whether the small liner moved before, with, or after the beam.

Investigator note

Common points: falling beam, no observed person contact, no observed crane movement. Non-common points: W-01 reports seeing initial motion; W-02 begins observing after a sound. Preserve both accounts and their visibility limits.

Crane Activity and Lift Check

A synthetic operational log used to test whether crane interaction is supported, contradicted, or merely unobserved.

Document ID	Incident	Record type	Status
R3-EV-010	R3 carrier beam - 10/14/2024	Application test evidence	Synthetic / controlled

Relevant interval

Time window	Logged activity	Corroboration	Status
11:30 AM-12:10 PM	Crane assigned to a different bay; no R3 lift ticket.	Dispatch log and operator entry	Synthetic scenario fact
12:10 PM-1:10 PM	No commanded crane travel or hoist movement in R3 bay.	Controller event log	Synthetic scenario fact
Approx. 1:03 PM	No active hook, rigging, or suspended crane load at R3.	Witnesses and controller log	Consistent
Prior shift roll removal	Crane-assisted roll-removal activities completed before the relevant interval.	Work log; exact final-clear time recorded separately	Historical source fact / detail synthetic

Boundary

This log addresses crane movement and lifts. It does not exclude earlier contact, rigging interference, or a configuration change unless those possibilities are checked against the roll-removal records and physical evidence.

Hydraulic Status and Bleed Verification

A synthetic field-data summary distinguishing recorded pressure state from mechanical support state.

Document ID	Incident	Record type	Status
R3-EV-011	R3 carrier beam - 10/14/2024	Application test evidence	Synthetic / controlled

Recorded readings

Point / source	Reading or state	Time	Limitation	Status
B-system supply gauge	0 psi after bleed	10/14 - 12:28 AM	Local gauge accuracy not independently calibrated for investigation.	Synthetic scenario fact
B-system return / bleed	Bleed path recorded open	10/14 - 12:28 AM	Exact trapped volumes not mapped in field record.	Synthetic scenario fact
Top balance position	Raised	Turnover entry	Position does not establish the load path after pressure removal.	Historical source fact
Pressure trend	No historian tag identified for isolated branch.	Relevant interval	Cannot reconstruct transient pressure from trend data.	Not available
Leak-down test	Not performed before post-event disturbance.	After event	No direct rate measurement exists.	Not completed

Interpretive boundary

The record supports that B hydraulics were intended to be at zero pressure. It does not establish how the raised balance and carrier assembly were mechanically supported, nor whether movement occurred during pressure decay.

Scene Photograph Index and Measurement Log

A synthetic visual-evidence index. Graphic panels are placeholders, not recreations of actual photographs.

Document ID	Incident	Record type	Status
R3-EV-012	R3 carrier beam - 10/14/2024	Application test evidence	Synthetic / controlled

P01

Overall landing area

Beam on sled / scaffold

Synthetic image placeholder

P02

Link and keeper

Keeper visible as installed

Synthetic image placeholder

P03

Yoke pocket

No gross fracture visible

Synthetic image placeholder

P04

Window liner

Recovered with beam

Synthetic image placeholder

P05

Contact surfaces

As-found marks recorded

Synthetic image placeholder

P06

Upper stand

View partly obstructed

Synthetic image placeholder

Measurement log

Item	As-found observation	Measurement status	Status
Carrier beam	No gross fracture visible; landed across sled and scaffold platform.	Overall dimensions not re-measured before recovery.	Historical source fact / measurement open
Link	Recovered with beam and described as intact.	Critical wear and seating dimensions pending controlled inspection.	Historical source fact / pending
Keeper	Present and installed on recovered assembly.	Fit, deformation, and retained-pin clearance pending.	Historical source fact / pending
Yoke	No apparent gross damage in initial view.	Pocket geometry and contact-surface measurements pending.	Historical source fact / pending
Window liner	Small top-operator-side liner recovered with beam.	Pre-event location and attachment condition not documented in starter record.	Historical source fact / open

Original-file requirement

For a real investigation, upload the original image files with EXIF metadata and the scale/measurement method. This synthetic index is designed to prompt that request; it is not a substitute for imagery.

Post-Event Mechanical Inspection

A synthetic condition report separating gross visual findings from dimensional and functional conclusions.

Document ID	Incident	Record type	Status
R3-EV-013	R3 carrier beam - 10/14/2024	Application test evidence	Synthetic / controlled

Inspection findings

Component	Visual finding	What is not established	Status
Carrier beam	No visible fracture or permanent gross bend in preliminary visual check.	Load history, subtle distortion, and material condition.	Synthetic scenario fact
Link	No visible fracture; one contact edge shows a polished witness area.	When the mark formed and whether it indicates partial seating.	Synthetic scenario fact / interpretation open
Keeper and pin	Keeper present; pin retained.	Keeper engagement before the fall and dimensional condition.	Historical source fact / details open
Yoke pocket	No gross crack or break; contact surfaces require cleaning and measurement.	Wear, taper, clearance, and dynamic path.	Historical source fact / pending
Window liner	Recovered displaced; attachment hardware condition not fully documented.	Whether liner movement preceded or followed beam motion.	Open

Recommended follow-up tests

- Controlled dimensional inspection against the full drawing and tolerance stack.
- Material and non-destructive examination if engineering identifies critical areas.
- Fit-up demonstration using an exemplar or digital model without disturbing original evidence.
- Failure-mode review distinguishing fracture, pin release, link unseating, interference, and support loss.

Similar-Stand Comparison - R1, R2, R4, and R5

A synthetic comparison record based on the immediate action to inspect other stands. It provides context, not proof of R3's pre-event state.

Document ID	Incident	Record type	Status
R3-EV-014	R3 carrier beam - 10/14/2024	Application test evidence	Synthetic / controlled

Comparison results

Stand	Keeper present	Balance position at inspection	Visual link seating	Disposition
R1	Yes	Lowered	Centered in visible yoke pocket	No gross anomaly
R2	Yes	Lowered	Centered; contact surfaces not measured	No gross anomaly
R4	Yes	Lowered	Slight visual asymmetry; engineering measurement requested	Follow-up
R5	Yes	Lowered	Centered in visible yoke pocket	No gross anomaly

Comparison limits

- All other balances were lowered before or during the immediate safety response.
- The comparison does not reproduce R3's raised, roll-removed, hydraulically bled configuration.
- Visual similarity is not a tolerance check.
- The R4 observation is an inspection lead, not evidence that it shares the R3 event mechanism.

Scaffold Inspection and Mill-Pit Occupancy Record

A synthetic access record used to establish exposure, last-known occupancy, and post-impact scaffold status.

Document ID	Incident	Record type	Status
R3-EV-015	R3 carrier beam - 10/14/2024	Application test evidence	Synthetic / controlled

Access and inspection log

Time	Entry	Record source	Status
6:20 AM	R3 scaffold inspected and tagged available for planned configuration.	Competent-person scaffold tag	Synthetic scenario fact
6:45 AM-12:35 PM	Contractor crew entries and exits recorded intermittently for pit work.	Pit access sheet	Synthetic scenario fact
12:42 PM	Last recorded contractor exit before event.	Pit access sheet	Synthetic scenario fact
Approx. 1:03 PM	No personnel in pit when carrier beam fell.	Preliminary report and witnesses	Historical source fact
After event	Platform quarantined; impact damage assessment required before reuse.	Area-control log	Synthetic scenario fact

Exposure statement

The absence of personnel at the moment of impact prevented injury. Earlier occupancy beneath or near the potential fall path remains relevant to potential severity and control adequacy, but it does not show that worker action caused the fall.

Contractor Crew Roster and Qualification Check

A synthetic anonymized qualification record. It supports competence and exposure checks without supplying personal data or implying misconduct.

Document ID	Incident	Record type	Status
R3-EV-016	R3 carrier beam - 10/14/2024	Application test evidence	Synthetic / controlled

Anonymized roster

Worker ID	Assigned role	Relevant documented qualification	Pit access on 10/14	Status
C-101	Mechanical lead	Site orientation; energy-control affected-person; task qualification current.	Morning	Current
C-102	Mechanical craft	Site orientation; scaffold-user; task qualification current.	Morning	Current
C-103	Mechanical craft	Site orientation; scaffold-user; task qualification current.	Morning	Current
C-104	Helper / spotter	Site orientation; task-specific briefing documented.	Morning	Current
C-201	Witness / nearby crew	Site orientation current.	Not in pit at event	Current
C-202	Witness / nearby crew	Site orientation current.	Not in pit at event	Current

Record limits

- This check establishes only the listed training records; it does not prove task execution quality.
- No person is identified as having touched or altered the carrier assembly before the event.
- Authorization for energy-isolation leadership must be checked separately from affected-person training.

Preserved Video Review Log

A synthetic review of a partial camera view. It tests temporal corroboration without pretending that hidden geometry is visible.

Document ID	Incident	Record type	Status
R3-EV-017	R3 carrier beam - 10/14/2024	Application test evidence	Synthetic / controlled

Camera and preservation

Camera	Coverage	Preserved interval	Limitation
Roughing-bay fixed camera - synthetic	Walking route, part of R3 stand, lower pit opening	12:50 PM-1:10 PM	Upper yoke/link interface obscured by stand structure; no useful audio

Observed sequence

Timestamp	Observation	Confidence	Status
1:03:09 PM	No person or crane hook visible within the immediate carrier-beam area.	High within camera field	Synthetic scenario fact
1:03:11 PM	Lower end of beam begins a small visible movement.	Moderate; upper connection hidden	Synthetic scenario fact
1:03:12 PM	Beam rotates and exits the visible support position.	High	Synthetic scenario fact
1:03:13 PM	Beam enters pit; dust / motion briefly obscures landing area.	High	Synthetic scenario fact
1:03:20 PM	Nearby personnel stop and move away from the stand.	High	Synthetic scenario fact

What video cannot answer

- Whether the link was fully seated in the yoke before motion began.
- Whether hydraulic or mechanical movement occurred inside the obscured upper assembly.
- Whether the small liner moved first.
- Exact fall geometry or component clearances.

OEM / Engineering Information Request

An intentionally unresolved evidence request. It tests whether the investigation can keep a causal branch open while awaiting authoritative design information.

Document ID	Incident	Record type	Status
R3-EV-018	R3 carrier beam - 10/14/2024	Application test evidence	Synthetic / controlled

Questions sent for authoritative response

Request	Why needed	Current status
Full carrier-beam, link, keeper, and yoke assembly drawing with revision history	Establish design intent, geometry, and positive-retention features.	Pending
Permitted balance positions during roll removal and hydraulic bleed-down	Determine whether the observed raised/isolation state was within approved configuration.	Pending
Required temporary blocking or restraint	Compare field record with manufacturer-defined gravity control.	Pending
Inspection limits for yoke and link contact surfaces	Evaluate wear and partial-seating hypotheses.	Pending
Known service bulletins or comparable events	Assess whether a latent design or known-condition pattern exists.	Pending

Investigation handling

Do not convert these requests into facts. Causes that depend on an OEM answer should remain provisional, with the missing evidence explicitly attached to the branch. Other supported branches may continue in parallel.